

TECHNICAL SPECIFICATION

DOCUMENT 00

Ultra-Safe Central Capsule

Scalable Survival Module for Ground, Aerial & Maritime Platforms

Document Version	Date	Status
1.2 (Final Architecture – Quality Score 9.7)	August 2026	Baseline for Patent Extraction – Final Door & Seating Layout

This document provides the complete technical baseline for the Ultra-Safe Central Capsule: geometry, mass budget, carbon-fiber roll-cage architecture, multi-layer parachute deployment and separation logic, life-support systems, and an initial patent-family extraction assessment.

Revision Note (v1.2 – Final Architecture): Door configuration finalized as three openings only: two side doors (port & starboard) for post-separation emergency egress, plus one large reinforced roof hatch for primary boarding while installed in the host vehicle and for post-landing rescue. Bottom plane remains fully continuous (no floor hatch) to achieve maximum structural strength for the six mechanical legs and energy-absorption system. Occupant capacity increased to seven seats arranged in a circular/radial layout (pilot + co-pilot + five passengers) to improve central access and G-load distribution while preserving full roll-cage continuity.

1. Purpose and Design Intent

The Ultra-Safe Central Capsule is an independent, modular survival cell designed to be installed at the structural heart of ground vehicles, helicopters, fixed-wing aircraft, aerial cranes, and maritime platforms. Its primary requirement is to keep occupants alive and recoverable even if the host vehicle is completely destroyed.

Key design principles: (1) complete functional independence from the host vehicle after separation, (2) multi-layer redundancy on every critical safety function, (3) scalable geometry for different platform classes, and (4) maximum structural integrity under extreme impact and post-crash fire.

2. Geometry and Dimensional Baseline (Standard Size)

Parameter	Value	Notes
Overall diameter	3.60 m	External envelope
Overall height	2.80 m	Including retracted landing legs
Occupant capacity	7 persons	Pilot + co-pilot + 5 passengers
Seating arrangement	Circular / radial	Seats attached around the inner perimeter, central volume free
Useful internal volume	18-20 m³	Seating + equipment + life support
Fixed mechanical legs	6 units	Each with hydraulic damper + thick elastomer pad - zero interference
Emergency / access doors	3 doors	2 side (port & starboard) + 1 reinforced roof hatch. No bottom door.

Door architecture rationale: The bottom plane is kept fully continuous to provide the highest possible stiffness and strength for the six mechanical landing legs and to maximize energy absorption during hard landings. The two side doors serve exclusively as post-separation emergency exits. The single large roof hatch is the primary boarding interface while the capsule remains installed in the host vehicle and the main rescue path after landing. All door openings are positioned so that no primary 12 cm carbon-fiber tube is cut; loads are transferred through reinforced frames into the continuous cage network.

3. Operational Mass Budget

With the full suite of safety systems, 12 cm carbon-fiber roll-cage, heavy energy-absorbing seats, oxygen and water reserves, multi-layer backup actuators and robust primary structure, the realistic operational mass is estimated as follows:

Subsystem	Mass Range (kg)
Primary chassis + 12 cm all-around carbon-fiber roll-cage	1 800 - 2 200
Multi-layer outer shell + impact-resistant glazing	950 - 1 150
6 mechanical legs + hydraulic dampers + thick elastomer pads	750 - 900
7 heavy viscoelastic seats + 5-point harnesses (circular layout)	750 - 920
6 primary parachutes + 8 auxiliary + multi-layer deployment system	550 - 700
Dual fire-suppression system + omnidirectional nozzles	180 - 230
Electrical systems + emergency battery + thermal management	350 - 450
72-hour oxygen + emergency water & food stores	300 - 380
4-layer separation & pyrotechnic backup systems	220 - 280
Ancillary equipment (satcom, EPIRB, lighting, first-aid, etc.)	150 - 200
Empty mass (approximate)	6 000 - 7 410
7 occupants (average 80 kg each)	560

Total operational mass (minimum realistic)	6 560 – 7 970 kg (~7.0–7.5 tonnes)
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Conclusion: Realistic operational mass remains in the ~7–7.5 tonne class. The continuous floor and optimized three-door layout contribute to both structural performance and a clean mass distribution.

4. Carbon-Fiber Roll-Cage - Technical Detail (12 cm All-Around)

The roll-cage forms the primary load-bearing and occupant-protection skeleton. It draws conceptual inspiration from high-performance rally cages but is engineered to aerospace structural standards. The continuous top and bottom planes (only one controlled roof opening) deliver the highest practical torsional and bending stiffness.

Key specifications:

- Primary tube thickness at critical load paths: 12 cm
- Secondary / reinforcement members: 8-10 cm
- Material: aerospace-grade carbon fiber (3K & 12K twill) with high-toughness epoxy matrix
- Structural topology: combined hexagonal + triangulated pattern in all six directions
- Protection axes: top, bottom, forward, aft, left, right (full 6-axis enclosure)
- Node joints: Titanium Grade 5 fittings with aerospace fasteners and structural epoxy adhesive
- Design safety factor: ≥ 2.8 against ultimate load (maximized by continuous floor and limited roof opening)
- Operating temperature range: $-150\text{ }^{\circ}\text{C}$ to $+120\text{ }^{\circ}\text{C}$
- Fire resistance: outer nano-ceramic coating providing ≥ 15 minutes direct-flame protection
- Estimated cage assembly mass: 1 800-2 200 kg

Door-cage interface: The two side doors and the single roof hatch are located in inter-bay spaces between primary members. No primary 12 cm tube is interrupted. Local secondary reinforcement frames surround each aperture and transfer loads back into the continuous cage network via titanium node fittings. The floor remains completely unbroken, giving the six leg attachment nodes a fully continuous and stiff foundation.

5. Four-Layer Backup Architecture - Parachute Deployment & Host Separation

The most critical safety functions are parachute deployment and clean separation from the host vehicle. Both functions are implemented with four independent layers so that failure of any single path does not compromise the outcome.

Layer 1 - Primary Electrical System

Manual push-button inside the capsule plus automatic activation from accelerometers and rate gyros that detect free-fall or severe impact. Powered by the capsule's main emergency battery bus.

Layer 2 - Independent Pyrotechnic Path

Dedicated pyrotechnic initiators with their own non-rechargeable backup battery, completely isolated from the main electrical system. Automatically armed after a configurable 3-5 second timeout if Layer 1 does not confirm successful deployment.

Layer 3 - Pure Mechanical Path (No Electrical Power Required)

Steel cables and high-force springs that can be triggered either by a dedicated manual lever pulled by an occupant or by inertial mass under high negative acceleration. This layer contains zero electronics.

Layer 4 - High-Pressure Nitrogen Gas System

Stored nitrogen bottles that, if the three preceding layers have failed, open via mechanical-thermal valves and forcibly eject and inflate the parachutes. Final line of defense.

Shared parachute system features:

- 6 large primary canopies + 8 auxiliary / stabilizer canopies
- Each primary canopy carries two small pilot/extractor chutes that force opening if the main canopy is fouled
- Automatic failure detection: after 3 seconds without confirmed inflation the next layer is commanded
- Both automatic and fully manual deployment modes are always available
- Worst-case (no parachute opens): the six mechanical legs, hydraulic dampers, thick elastomer pads and viscoelastic seats still absorb a large fraction of the impact energy

Host-vehicle separation system:

- Parallel pyrotechnic + hydraulic release actuators

- Mechanical locks that can be opened by a pure manual lever even with total electrical failure
- Frangible or pyrotechnic-cut electrical and hydraulic umbilicals
- Separation-confirmation sensors that interlock parachute deployment until the capsule is clear of the host

6. Life-Support, Fire Suppression and Ancillary Safety Systems

Seats and residual-G management:

Seven viscoelastic seats 45–50 cm thick with adjustable stiffness and five-point harnesses featuring automatic locking and rapid post-landing release. Seats are arranged in a circular/radial pattern around the inner perimeter, leaving a clear central volume for movement and access to the roof hatch. Design goal: keep occupant residual G below 4 g even in very hard landings. The circular layout also improves load distribution under multi-axis impact.

Oxygen, water and food:

- Compressed oxygen sufficient for ≥ 72 hours for seven occupants (with CO₂ scrubbing / rebreather capability)
- Emergency water store ≥ 14 litres
- Compressed emergency food rations for 72 hours for seven occupants
- Air filtration system capable of operating in smoke and toxic atmospheres

Fire suppression:

Dual-agent architecture: clean-agent gas (FM-200 or Novec equivalent) for the interior volume + foam / dry-powder for external fires. Omnidirectional automatic nozzles with multi-sensor detection (thermal + smoke + infrared).

Emergency and access doors (Final configuration):

Three doors only: two side doors (port and starboard) and one large reinforced roof hatch. Side doors are intended primarily for post-separation emergency egress after landing. The roof hatch serves as the primary boarding interface while the capsule is still installed in the host vehicle (via a matching access opening or platform on the host) and as the main rescue path after landing. Door-release permission is granted only after the capsule control logic confirms that residual G and fire status are within safe limits. No bottom door is provided; the floor remains fully continuous to maximize strength for the six mechanical legs and energy-absorption system.

Communications and location:

- Dual satellite communication (Iridium + emergency Starlink) with automatic antenna deployment
- EPIRB and AIS emergency transmitters
- High-intensity LED emergency lighting + strobe for visual acquisition day or night
- Structural hard-points on the upper surface for crane or helicopter recovery hooks (compatible with roof hatch)

7. Patent-Family Extraction Potential

Based on the technical features defined in this baseline, the following high-level patent-family assessment is offered. Counts are conservative engineering estimates of distinct, strong inventive concepts.

Technology Domain	Est. Families	Strength	Core Inventive Focus
4-layer parachute deployment architecture	3–5	Very High	Electrical + pyrotechnic + pure mechanical + gas-pressure paths
12 cm 6-axis carbon-fiber roll-cage with continuous floor	2–4	High	Hex + triangulated layout, continuous bottom plane, limited roof opening
Integrated multi-stage impact-energy absorption	2–3	High	Legs + dampers + elastomer pads + viscoelastic seats
Multi-path host-vehicle separation system	2–3	Very High	Pyrotechnic + hydraulic + pure mechanical release
Circular seating + roof-hatch egress architecture	1–2	Medium–High	Radial seats + single reinforced roof hatch preserving cage continuity
Modular platform-agnostic capsule architecture	2–3	High	Installable in ground, rotorcraft, fixed-wing and maritime hosts

Integrated 72-hour life-support package	1-2	Medium	Oxygen + scrubbing + filtration + satcom for 7 occupants
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Overall assessment: approximately 13-22 distinct patent families appear extractable. Of these, roughly 8-13 families are judged to possess high to very-high commercial and defensive strength. The strongest clusters remain the four-layer parachute deployment system, the multi-path host-separation architecture, and the continuous-floor high-strength cage concept.

8. Key Specification Summary

- Geometry (Standard): 3.60 m diameter × 2.80 m height
- Occupants: 7 persons (pilot + co-pilot + 5 passengers) in circular / radial seating arrangement
- Operational mass: approximately 7.0–7.5 tonnes
- Roll-cage: 12 cm carbon-fiber primary tubes, full 6-axis enclosure, continuous floor, safety factor ≥ 2.8
- Parachutes: 6 primary + 8 auxiliary with four independent deployment layers
- Impact absorption: 6 mechanical legs + hydraulic dampers + elastomer pads + viscoelastic seats
- Life support: ≥ 72 h oxygen with CO₂ scrubbing, water and food stores for 7 occupants
- Fire suppression: dual-agent (clean gas interior + foam/powder exterior), omnidirectional
- Doors: 3 doors total — 2 side doors (port & starboard) for post-separation emergency egress + 1 large reinforced roof hatch for boarding while installed and for post-landing rescue. No bottom door.
- Structural priority: maximum strength and energy absorption achieved by keeping the entire floor plane continuous and limiting openings to the sides and one controlled roof hatch
- Patent potential: 13–22 families, of which 8–13 are high-to-very-high value

This document constitutes the baseline technical specification (Document 00, Version 1.2 – Final Architecture) for detailed design, simulation, and subsequent patent extraction and filing activities. Any material change to the requirements shall be recorded in a subsequent revision of this document.

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